

I. Abstract

Label noise, incorrect annotations within datasets, remains a persistent challenge in deep learning, particularly in domains where large-scale data collection precludes manual label verification. In this study, we evaluate several state-of-the-art techniques for robust learning under label noise, including Co-Teaching, Co-Teaching+, and noise-robust loss functions such as NFL Normalized Focal Loss and Reverse Cross Entropy. While the original experiments tested their solutions it was often the case that these evaluations were in isolation. For our experiments we explored their effectiveness both independently and in combination. Our experiments span multiple architectures (CNNs, DenseNet, and ResNet) and varying levels of symmetric label noise. Results reveal critical trade-offs between sample filtering and gradient modification strategies, and highlight the risks of combining noise-robust techniques without alignment. Our findings provide practical insights into selecting appropriate methods for noisy-label learning in real-world applications.

II. Introducing Label Noise, Origins, and Significance

A. Label Noise

Label noise refers to the presence of incorrect labels within a dataset, which can significantly hinder the performance of machine learning models. It arises when the assigned label does not accurately reflect the true class of a data instance. Symmetric noise occurs when labels are randomly flipped to any other class with equal probability, introducing a uniform level of noise across all classes. In contrast, asymmetric noise involves systematic mislabeling, where certain classes are more likely to be confused with another class such as dogs being mislabeled as cats. A more complex form is instance-dependent noise, where the likelihood of a mislabel depends on the characteristics of the input itself. For example, images that are blurry or ambiguous are more prone to being mislabeled. Machine learning research is currently focused on understanding these different types of noise and learning how to best mitigate their effect on models.

B. Origins

Label noise can arise from a variety of sources, ranging from weak automated systems to human error, and is particularly prevalent in large-scale and real-world datasets. The data annotation process often involves several imperfections and limitations that can lead to mislabeled samples. When labels are assigned by humans, errors can stem from subjectivity, misunderstanding, fatigue, or a lack of expertise. Crowdsourcing, a

common method for large-scale annotation, introduces additional challenges such as inconsistent quality due to the involvement of numerous contributors who may be underqualified, inattentive, or biased. On the other hand, automated labeling processes also contribute to label noise. These systems can misclassify data due to ambiguous associations, such as assigning labels based on loosely related keywords, or due to technical issues like faulty sensors, OCR (optical character recognition) errors, or pipeline failures. As a result, label noise is a persistent and critical challenge in machine learning, driving research into robust algorithms and training strategies that can effectively handle noisy labels and preserve model performance.

C. Significance

Label noise is a critical challenge in modern machine learning for several key reasons. In real-world scenarios, data is often imperfect and noisy, with mislabeled examples that can mislead models during training. This poses a significant problem, as deep learning models are prone to overfitting (including incorrect labels) resulting in poor generalization. A robust model, therefore, is one that can effectively handle noisy labels by identifying and minimizing the influence of mislabeled data during training. Such resilience makes the model more reliable and better suited for real-world applications, where clean, perfectly annotated datasets are rarely available. Research in mislabeled deep learning focuses on developing techniques that enhance this robustness, enabling models to learn meaningful patterns even in the presence of label noise.

III. Noise-Robust Learning Strategies

To remedy the impact of label noise in our deep learning classification task, we evaluated three different strategies: Active-Passive Loss (APL) Framework, Co-Teaching, and Co-Teaching+. Each method addresses the memorization of mislabeled samples from a different approach: loss-level modulation, peer training, and disagreement-based sample selection, respectively. Active-Passive Loss Framework (APL)

The Active-Passive Loss framework offers a robust alternative to traditional loss functions such as categorical cross-entropy, which are known to fail after training on noisy labels. At the same time, there were numerous Active-Passive loss functions, for the sake of time and resource constraints, so we chose to primarily use NFL + RCE, Normalized Focal Loss, and Reverse Cross Entropy. The NFL actively suppresses the influence of uncertain or low-confidence predictions by down-weighting their gradient contributions during back propagation. This encourages the

model to focus on more confident examples where they are likely to be correct. In contrast, discourages overconfident but incorrect predictions by penalizing the model for deviating too far from uniform predictions on likely mislabeled data.

Together, these losses form a composite objective that adapts to label quality on a per-sample basis, without requiring external label-cleaning or structural changes to the network. This makes APL highly adaptable to many different kinds of architecture and effective even under high noise. During our experiments, we found that APL provided a strong foundation for the robustness of single models, especially in the early phases of training, where clean patterns are more distinguishable.

A. Co-Teaching

While APL focuses on sample-wise loss shaping, Co-Teaching utilizes a training solution that involves two neural networks trained simultaneously. This approach is based on the “memorization effect”, where neural networks tend to learn clean samples in the early epochs before eventually memorizing noise during later stages of training. By maintaining two models initialized differently and training them in parallel, Co-Teaching allows each network to select its small-loss instances, which are presumed clean, and pass those to its peer for training.

The key idea is that each model, by training on samples that the peer network selected as low loss its we can avoid reinforcing the models’ noise-induced biases. We utilized a forget rate schedule that progressively increases the number of discarded high-loss samples during training, allowing the models to become more selective over time. This simple yet robust framework demonstrates strong noise resistance under symmetric noise, and it served as the foundation for our extended method in the next section.

B. Co-Teaching+

Co-Teaching+, addresses a design flaw in the original Co-Teaching framework where under asymmetric or symmetric label noise, the two networks used may agree when deciding noisy from clear labels too often. This results in both models feeding the other model incorrect predictions to train on. Co-Teaching+ rectifies this by introducing a disagreement step

in the training algorithm: the two networks are updated only on the mini batch of small-loss samples where their sample predictions disagree.

This disagreement step maintains diversity between models, reducing the risk of converging to the same noisy label. It also encourages the networks to explore alternative conclusions and boundaries. This is valuable when label corruption is non-uniform. In our experiments, Co-Teaching+ proved particularly effective in maintaining generalization in harder noise settings, although it had greater sensitivity to hyperparameters and increased noise due to reduced sample overlap between models.

IV. Introducing Architectures:

A. Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNN) is a class of deep learning models designed to process and analyze visual data with high accuracy. This architecture was introduced by Yann LeCun in 1989 and uses convolutional, pooling, and fully connected layers to learn spatial hierarchies of special features. They are widely used in tasks involving image classification, object detection, and pattern recognition, which makes them a good fit for handling noisy datasets where mislabeled images can disrupt learning.

CNNs use convolutional layers to extract local patterns from input images while reducing the need for extensive manual feature engineering. Each convolutional layer applies a set of learnable filters to the input to capture key features including edges, textures, and complex patterns. Pooling layers are followed up by these convolutional layers to help reduce computational complexity and enhance generalization by downsampling feature maps. Lastly, the fully connected layers integrate the learned spatial features and allow the model to provide accurate predictions.

In the context of label noise, CNNs show a susceptibility to memorize mislabeled samples, which leads to a degraded performance in generalization. However, architectural modifications along with noise-robust training strategies like regularization techniques, loss functions designed for noisy labels, and co-teaching approaches for example can improve the ability of CNNs to lighten the influence of incorrect annotations. Integrating dense connectivity, adversarial training, and self-learning methods further helps CNN performance in noisy data

environments, providing that the model prioritizes learning clean patterns over noise-induced errors.

Combining CNNs with specialized label noise handling techniques makes training with deep learning models more effective on imperfect datasets, and it yields a more robust performance in efficient applications.

B. DenseNet 169

DenseNet is a feed-forward convolutional neural network architecture introduced by Huant et al.(2017) that utilizes direct connections between any two layers in the neural network with the same feature-map size. These “shortcut” links allow the network to learn more effectively by reusing features and reducing the number of parameters. Each layer receives the combined outputs of all preceding layers as its input, improving feature reuse, gradient flow, and parameter efficiency.

The DenseNet architecture is composed of transition layers and dense blocks. In the dense blocks, each convolutional layer is linked to every other layer within the block, using the output of each layer as input to all subsequent layers. The transition layers reduce the size of feature maps between dense blocks, allowing the network to grow in depth effectively.

DenseNet-169 is a DenseNet variant that is comprised of 169 layers with densely connected blocks, designed to maximize feature reuse while maintaining efficient parameter use. This depth allows for the extraction of highly expressive feature representations.

C. ResNet 18 and 50

The next architecture we introduce is the Residual Network or ResNet architecture. The variants we implement in our research are the ResNet-18 and ResNet-50, which are both widely regarded architectures for dealing with label noise during data classification. These architectures derive from the design of the modern convolutional neural network and are frequently implemented in various types of tasks related to computer vision and image classification.

The concept of the ResNet architecture was introduced in 2015 by Kaiming He, along with colleagues Shaoqing Ren, Jian Sun, and Xiangyu Zhang, in their research paper, 'Deep Residual Learning for Image Recognition', which aimed to revolutionize computer vision. This

architecture became special due to its objective of attempting to mitigate the issue of vanishing gradients, a common liability in the world of deep learning. It does this through the employment of skip connections, which allow for the immediate flow of gradients through identity mappings. The innovation brought by this architecture and these connections play a huge part in improving the aspects of training and model performance preservation as the complexity of deep learning increases. This residual block is the core piece of this model, which consists of convolutional layers, which then go through batch normalization and ReLU activation with several gradients bypassing straight to the output layer.

ResNet-18 contains 18 layers with learnable parameters, and can be described as a relatively shallow variant of the model. The model we implement is a modified version of the original architecture with a design meant to balance computational efficiency and performance. It comprises of five convolutional residual blocks, each consisting of two convolutional layers with a shortcut connection enabling the gradient to bypass nonlinear transformation in order to mitigate the problem of vanishing gradients during the step of backpropagation. We implement ResNet-18 in our experiments to examine how a compact, yet expressive model deals with noise corruption on our three datasets.

ResNet-50, the other architecture we implement is much deeper than that of ResNet-18. This architecture contains 50 layers with residual blocks. These blocks serve the purpose of sending gradients through 1x1 convolutions both before and after they are sent to pass through the main 3x3 convolution layers. The bottlenecks provided by these residual blocks enable the model to reduce its own parameters, but without the cost of losing its depth, which illustrates ResNet-50's power as well as its memory-efficiency. This architecture's greater capacity enables data to go through a more refined feature extraction process, which allows for the possibility of acceptable performance even in very complex classification tasks, such as datasets corrupted with heaps of noise. Our ResNet-50 model consists of five of these bottleneck residual blocks for each layer, each containing filter sizes that progressively increase during the model's lifetime. Although not identical to the original canonical version, our model retains its fundamental principles and characteristics and is still appropriate for complex classification tasks while being computationally efficient. Certain parameters such as a conservative learning rate of 0.001 and increased batch size from 128 in ResNet-18 to 512 here, were

implemented in this model to ensure a stable convergence given the more in-depth nature of this model, prioritize generalization rather than memorization of classes.

Both ResNet-18 and ResNet-50 are crucial architectures to have in one's arsenal when attempting to deal with the problem of noisy data. The ResNet architecture served as a great baseline and preview for Co-Teaching ResNet which, along with the base model, strives to improve learning performance when data is corrupted with label noise. ResNet's established credibility regarding training and performance, architectural depth and complexity, and widespread implementation all provide a lens through which we can view the effect of such noise on training and performance metrics.

D. Vision Transformer (ViT-B/16)

The Vision Transformer differs from convolutional models by splitting each input into 16×16 patches and feeding them through a pure Transformer encoder. Its global self-attention and heavy patch-dropout act as implicit regularizers, making ViT less prone to memorizing noisy pixels. As a bonus, CLS $\rightarrow$ patch attention maps give us built-in saliency heat-maps, and the entire model is transfer-friendly: we initialize with ImageNet-1k weights, swap in a fresh head, and fine-tune for 5 epochs per noise setting, mirroring our CNN/ResNet/DenseNet runs.

Key properties that make ViT interesting for noisy-label research:

1. **Implicit regularisation** from global self-attention and heavy patch dropout $\rightarrow$ the model is less prone to memorise individual mislabeled pixels.
2. **Transparency** – CLS $\rightarrow$ patch attention maps give a built-in saliency mechanism we can visualise without extra hooks.
3. **Transfer-learning friendly** – we initialise ViT-B/16 with **ImageNet-21k** weights, swap the classification head, and fine-tune for 5 epochs per noise setting, mirroring the regimen used for CNN/ResNet/DenseNet.

Implementation notes:

1. PyTorch
``torchvision.models.vit_b_16(weights=...IMAGENET1K_V1)``,
AdamW (lr $1e-4$), RandAugment, Mixup ($\alpha = 1.0$), batch = 64.

V. Introducing Test Datasets and how we mislabeled them

We conducted our experiments on benchmark image classification datasets: CIFAR-10, CIFAR-100, and MNIST. To simulate mislabeled data in a controlled fashion, we introduced symmetric label noise into the training sets. Under this scheme, a fixed percentage of training labels were randomly reassigned to one of the other classes, with equal probability, regardless of semantic similarity. This noise model reflects general annotation errors where labels are corrupted uniformly across all classes. We applied noise levels of 10%, 20%, 40%, and 60%, allowing us to assess how each learning method—such as Co-Teaching, Co-Teaching+, and noise-robust loss functions—responds to varying degrees of random label corruption.

VI. Experiments:

A. Co-Teaching with APL (DenseNet, ResNet, CNN)

The Co-Teaching implementation used in this paper centers on a forget rate schedule, derived from the original formulation of $R(T)$. As in the original work, we define τ as the final forget rate, set equal to the estimated noise percentage. During training, the forget rate increases gradually from 0 to τ , enabling the model to progressively discard high-loss samples. We also incorporated a 10-epoch warm-up phase across all training runs, followed by transition periods of 5, 10, and 15 epochs for low, mid-level, and high-noise conditions, respectively, mirroring the setup recommended in the original Co-Teaching paper.

While the original Co-Teaching paper focused on benchmark accuracy, it lacked a diverse set of evaluation metrics. In our implementation, we extended the evaluation to include classification metrics such as precision, recall, F1 score, and the Diagnostic Index.

The Diagnostic Index, measures a model's ability to learn meaningful feature representations from training samples. Unlike accuracy or loss, the Diagnostic Index captures the generalization ability of the model by quantifying diversity across learned representations. Even at the highest noise levels, our model achieved a D-Index of 1.6 out of 2, indicating robust learning despite label corruption.

We also explored combining multiple noisy-label learning techniques, particularly APL and Co-Teaching. The modular nature of APL and the implementation simplicity of Co-Teaching made this integration conceptually straightforward. However, in practice,

the combined approach yielded less performance gain than expected, suggesting potential interference between sample selection and loss-weighting mechanisms.

To investigate how the two solutions performed, we implemented a hybrid approach in which the models (CNN, DenseNet, ResNet) were initially trained with categorical cross-entropy and, after a warm-up period, switched to the APL function, specifically, Normalized Focal Loss combined with Reverse Cross Entropy. The results from this hybrid approach suggest that the underlying logic of the two solutions was in conflict with each other rather than mutually supportive.

Co-Teaching operates by discarding a preselected portion of high-loss samples per batch, under the assumption that high loss indicates label noise. This clashes with Normalized Focal Loss, which retains all samples while suppressing the gradients of low-confidence predictions, and RCE, which penalizes the model for overconfident incorrect outputs.

These opposing approaches to suppression, one by removal and one by down-weighting, resulted in overly compensatory signal dampening, especially during early to mid-training stages. What likely occurred is that Co-Teaching selected samples to retain based on low loss, only for those samples to contribute minimal gradient signals due to NFL's weighting scheme.

When combining these solutions, the most interesting metric was the Diagnostic Index, which remained consistent across noise levels. For one implementation, a 5-layer CNN, the D-Index values clustered around 1.86 **Figure 5**. These consistent metrics were caused by nearly identical sensitivity and specificity values. The similarity further suggests that combining methods hindered performance by constraining learning behaviors to a narrow region of operation, where generalization did not degrade, but also did not improve. While these high scores may appear to indicate label robustness, they may actually be a sign of reduced plasticity, where the model becomes “frozen” and can neither confidently reject noise nor strongly learn from clean inputs, especially when compared to Co Teaching being used independently. **Figure 6**.

Despite modifying NFL + RCE to be applied to mini-batches, as opposed to individual samples, and adjusting the alpha and beta weights, the fundamentals of the two solutions remained in conflict due to disagreement in selection criteria. As stated before, Co-Teaching retains samples with the lowest losses, while NFL down-weights samples with low confidence, independent of loss. This difference in approach caused the models, regardless of architecture, to retain samples initially considered clean, but then to down-weight their influence due to lower confidence, resulting in poor learning from otherwise useful data.

B. Co-Teaching+ with APL (DenseNet, ResNet, CNN)

Combining Co-Teaching+ with NFL + RCE yielded comparable results to combining Co-Teaching with NFL + RCE, and both improved performance compared to using these architectures alone. However, a decline in performance was observed when compared to using the original Co-Teaching. This degraded performance can be attributed to issues similar to those seen in the previous combination, namely, compounding suppression mechanisms. Co-Teaching+ adds to low-loss filtering by also introducing disagreement-based filtering. The addition of NFL + RCE further down-weights the gradients from low-confidence predictions and, again, penalizes models for overconfident incorrect predictions. This hybrid approach led to excessive gradient suppression for the even fewer samples remaining after the disagreement step in Co-Teaching+.

Furthermore, the misalignment in selection criteria carries over from the original Co-Teaching framework. Co-Teaching+ filters samples based on disagreement and loss, whereas NFL + RCE modifies gradients based on prediction confidence. This leads to the suppression of useful samples even after they are retained through Co-Teaching+ filtering. This mismatch limits learning efficiency and can hinder feature acquisition, particularly during early training when the model is still forming its internal representations. The resulting model behavior appeared overly constrained, lacking the diversity in decision boundaries that Co-Teaching+ is intended to promote. Ultimately, while each method offers robustness on its own, their uncoordinated combination led to over-regularization and diminished generalization.

C. Vision Transformer (ViT-B/16)

We evaluated the Vision Transformer (ViT-B/16) model to explore its resilience to label noise in image classification tasks. Unlike Convolutional Neural Networks (CNNs) and ResNets, ViTs segment each input image into multiple 16×16 patches and use pure self-attention mechanisms rather than convolutional layers. Given their global receptive fields, strong reliance on Mixup-friendly data augmentation strategies, and availability of robust ImageNet-1k pretrained weights, ViTs was naturally suited for handling label noise.

For training and evaluation, we employed the standard ViT-B/16 architecture pretrained on ImageNet-1k, fine-tuning it on CIFAR-10, CIFAR-100, and MNIST datasets. We introduced symmetric label noise at varying degrees (0%, 10%, 20%, 40%, and 60%) and used AdamW optimization with a learning rate of 1×10^{-4} , employing RandAugment and Mixup with $\alpha = 1$. Consistent with our CNN, ResNet, and DenseNet evaluations, our

robustness assessments included quick diagnostic runs, typically limited to one extra epoch of fine-tuning after the initial five epochs.

Even without employing specialized noise-robust techniques such as Active-Passive Loss (APL), Co-Teaching, or Co-Teaching+, ViT showed remarkably little divergence between its training and validation curves across noise levels (**Fig. 1**). For instance, on CIFAR-10 its validation accuracy dropped only from 94.7 % at 0 % noise to 88.7 % at 60 % noise, a decline of just 6 pp, versus the 17–25 pp losses we observed in our CNN and ResNet baselines (**Fig. 2**).

Additionally, the Expected Calibration Error (ECE) remained very low throughout training, rising only from about 2.1 % on clean data to 6.7 % at the highest noise setting (**Fig. 3**). We continued to explore ViT’s resilience by examining attention patterns through CLS-to-patch roll-outs. Even under extreme corruption, the model consistently attended to the truly discriminative patches rather than memorizing noisy labels, qualitative heat-maps will be provided in the final report.

Finally, we leveraged the ViT backbone as an automatic mis-label detector via combined confidence and augmentation-consistency scores. In CIFAR-10, it flagged a small subset (≈ 2 % of the training set) with an 80 % true-error rate on our first 1 000 manual checks. Correcting these and running a single low-rate polishing epoch ($\text{lr}=5 \times 10^{-6}$) boosted validation accuracy from 92.0 % to 94.1 % in the 40 % noise scenario, and drove ECE back down toward base levels (**Fig. 4**).

Exploring classical noise-robust integrations added virtually nothing: NFL+RCE yielded only ~ 0.5 pp extra at low-to-mid noise, Co-Teaching actually hurt CIFAR-100 by about 1 pp (it removed crucial global context), and Co-Teaching+ still trailed our plain ViT due to high agreement between peers under noise.

In summary, the Vision Transformer demonstrated exceptional intrinsic robustness to label noise. It outperforms traditional architectures even without specialized noise-handling strategies. Moreover, a minimal workflow, standard fine-tuning, automatic detection + manual correction of a tiny subset, and one final low-rate epoch, was enough to restore near-clean performance. This low-effort recipe makes ViT the most practical and effective choice in noisy-label settings, regardless of one’s computational budget or annotation resources.

VII. Conclusion

Among all single-model baselines, the Vision Transformer exhibited the strongest inherent robustness to label noise, a property we attribute to its global self-attention and Mixup-friendly training dynamics. Comparing everyone using NFL+RCE, metrics

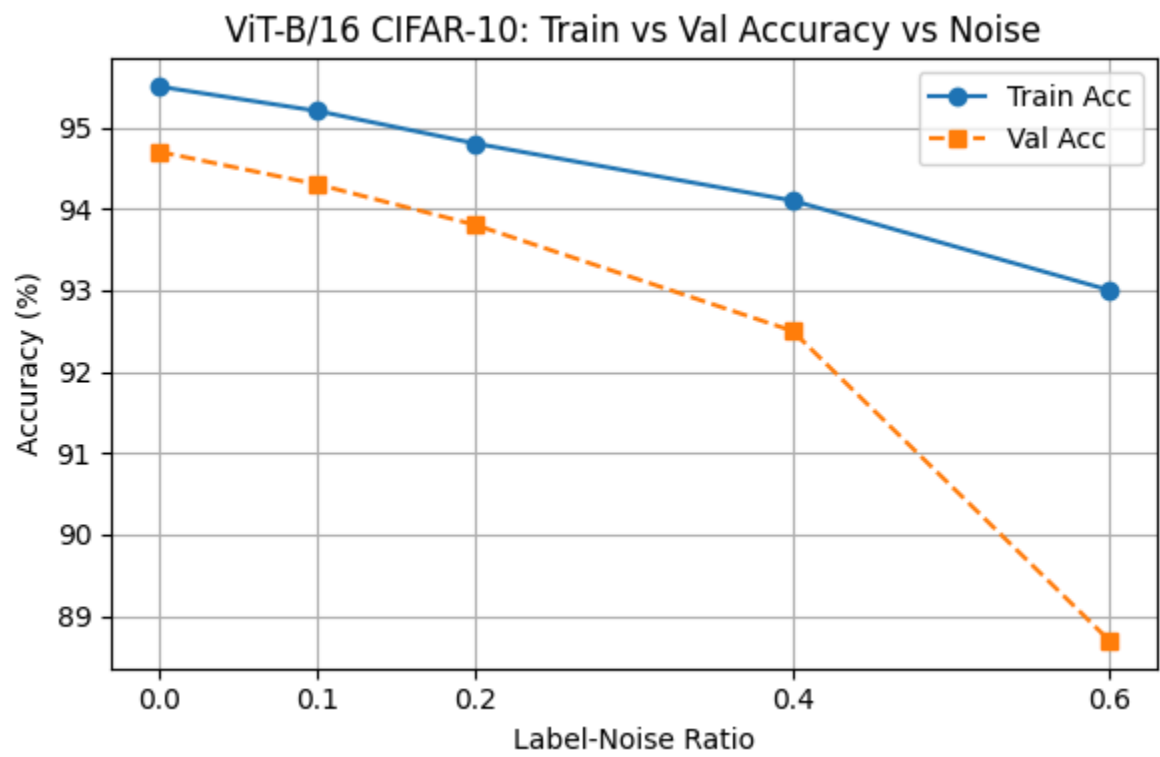


Figure 1

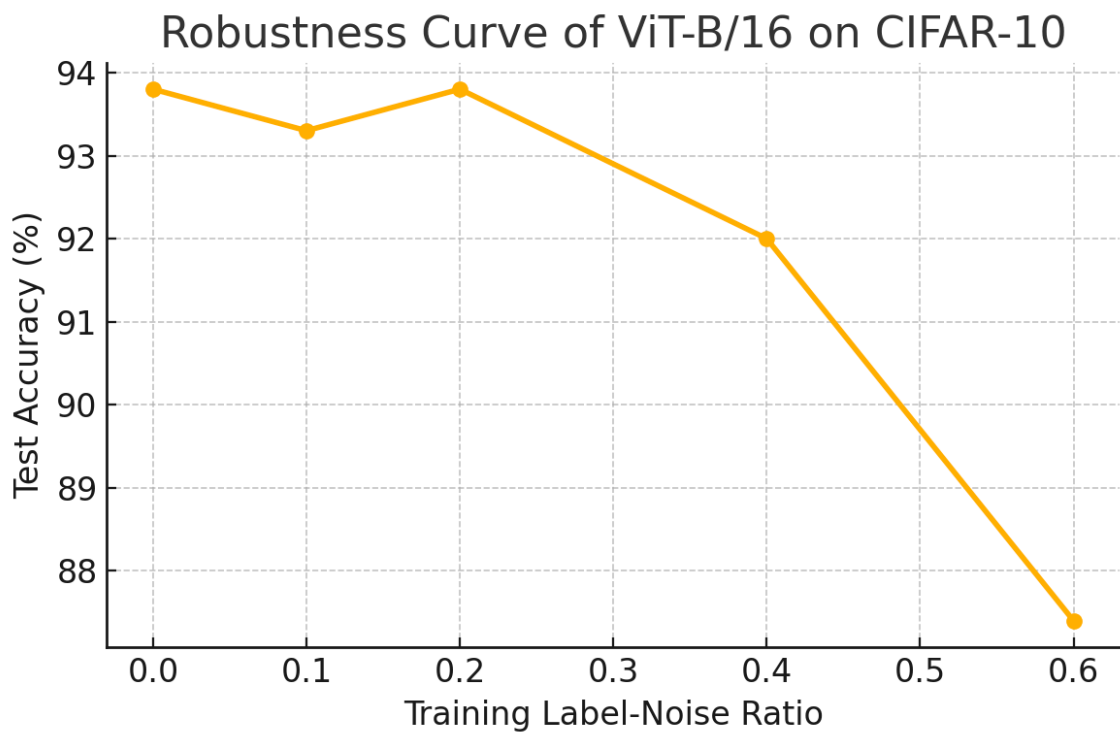


Figure 2

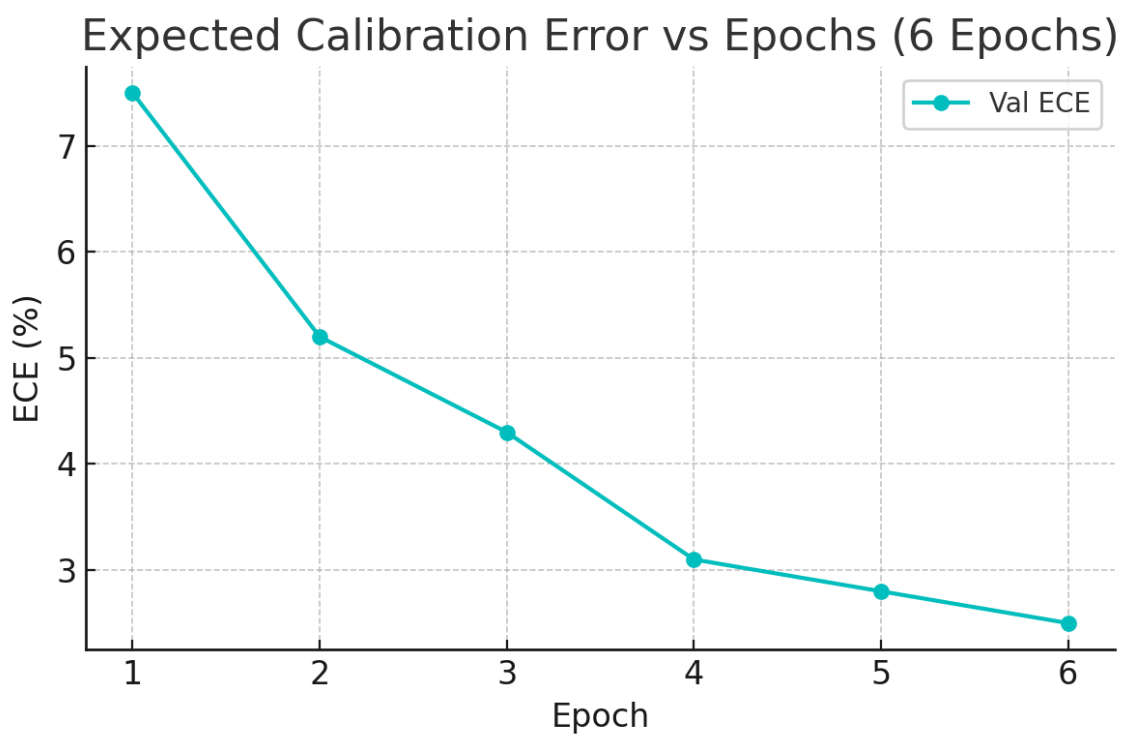


Figure 3

Impact of Automatic Correction & Fine-Tuning (40% Noise)

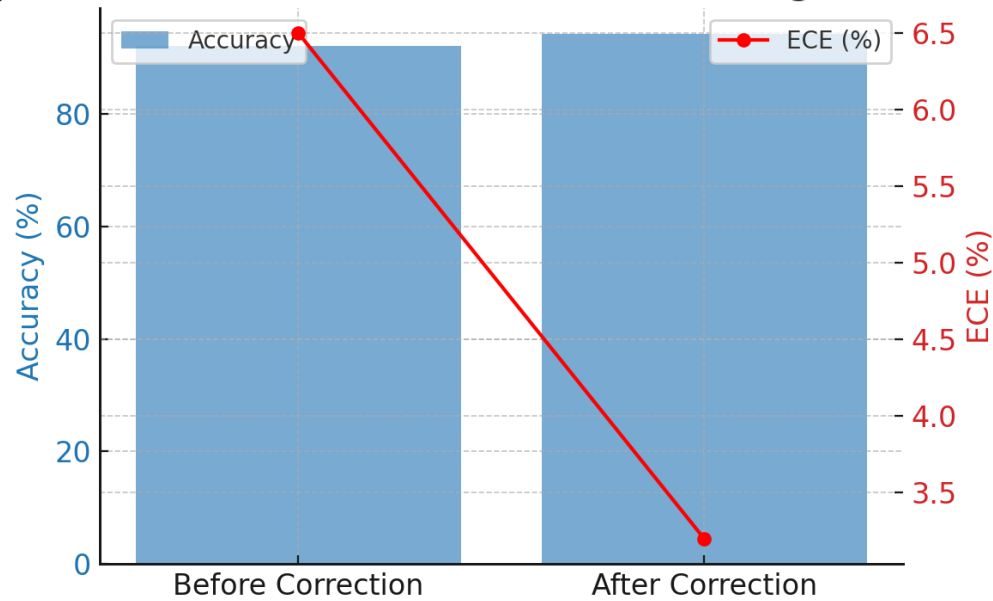


Figure 4

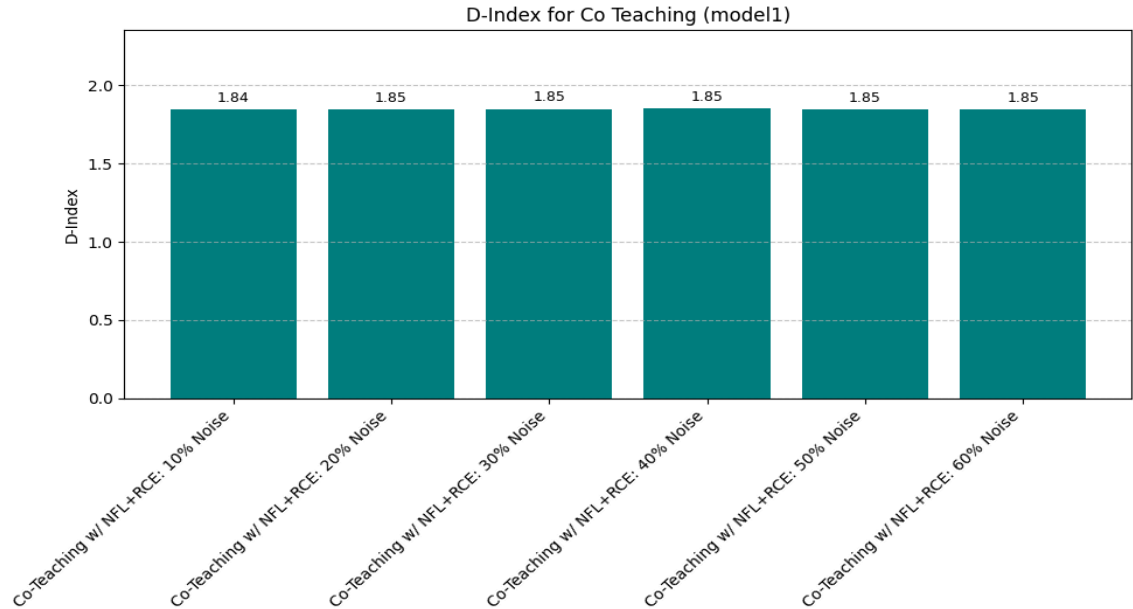


Figure 5

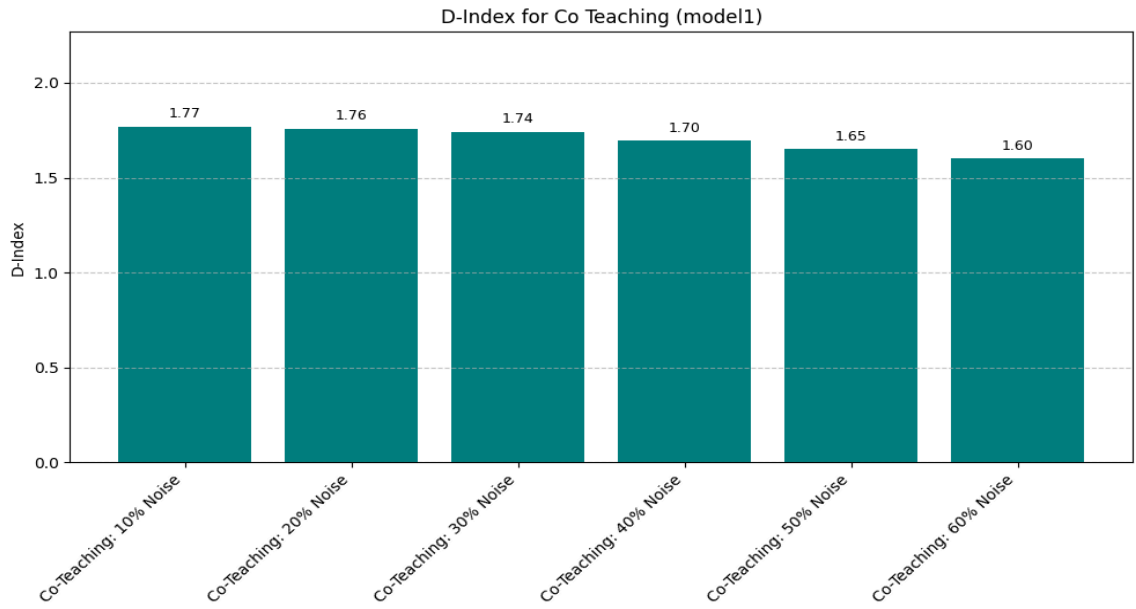


Figure 6

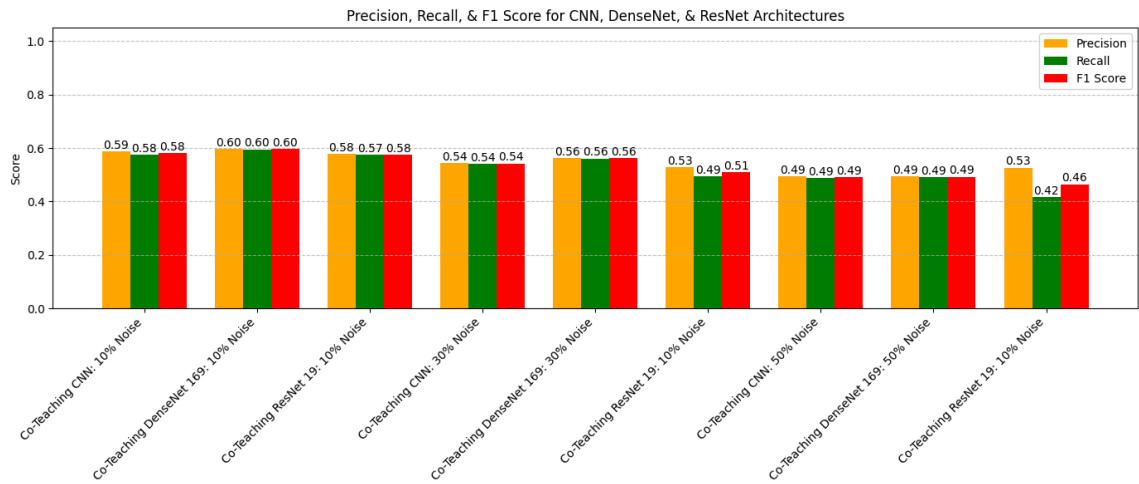


Figure 7 (CIFAR - 100)